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Monte Carlo sampling can be used to determine the size and shape of the steady-state flux space ☆

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Abstract

Constraint-based modeling results in a convex polytope that defines a solution space containing all possible steady-state flux distributions. The properties of this polytope have been studied extensively using linear programming to find the optimal flux distribution under various optimality conditions and convex analysis to define its extreme pathways (edges) and elementary modes. The work presented herein further studies the steady-state flux space by defining its hyper-volume. In low dimensions (i.e. for small sample networks), exact volume calculation algorithms were used. However, due to the #P-hard nature of the vertex enumeration and volume calculation problem in high dimensions, random Monte Carlo sampling was used to characterize the relative size of the solution space of the human red blood cell metabolic network. Distributions of the steady-state flux levels for each reaction in the metabolic network were generated to show the range of flux values for each reaction in the polytope. These results give insight into the shape of the high-dimensional solution space. The value of measuring uptake and secretion rates in shrinking the steady-state flux solution space is illustrated through singular value decomposition of the randomly sampled points. The V_{max} of various reactions in the network are varied to determine the sensitivity of the solution space to the maximum capacity constraints. The methods developed in this study are suitable for testing the implication of additional constraints on a metabolic network system and can be used to explore the effects of single nucleotide polymorphisms (SNPs) on network capabilities.
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Keywords: Constraint-based modeling; Extreme pathways; Monte Carlo; Hyper-volume; Flux cone

1. Introduction

Constraint-based modeling has been used to capture the phenotypic 'potential' of a metabolic network. The metabolic steady-state flux space, as defined by stoichiometric, thermodynamic, and reaction capacity constraints, describes a finite region of all attainable phenotypes (steady-state flux distributions) in the network (Schilling et al., 1999; Schuster et al., 1999; Beard et al., 2002). This solution space has been extensively explored using linear programming methods (Dean and Dervakos, 1998; Hatzimanikatis et al., 1998; Edwards et al., 1999; Schilling et al., 1999; Lee et al., 2000; Mahadevan et al., 2002; Segre et al., 2002). Constraint-based modeling involves the successive imposition of constraints so as to shrink the solution space and narrow the cell's operating range (Palsson, 2000). Recently, genetic regulation (Covert et al., 2001; Covert and Palsson, 2002, 2003) and experimentally measured metabolic flux data (Wiback et al., accepted) have been used to further shrink the steady-state flux space. A metric for measuring the size of this space and its reduction with additional constraints would be useful for evaluating various types of constraints and determining the importance of different parameters in the system.

Without upper bound constraints (V_{max}), the steady-state flux space is a convex cone (Fig. 1A). The V_{max} values of the enzymes in a metabolic network put an upper limit on the amount of flux able to pass through

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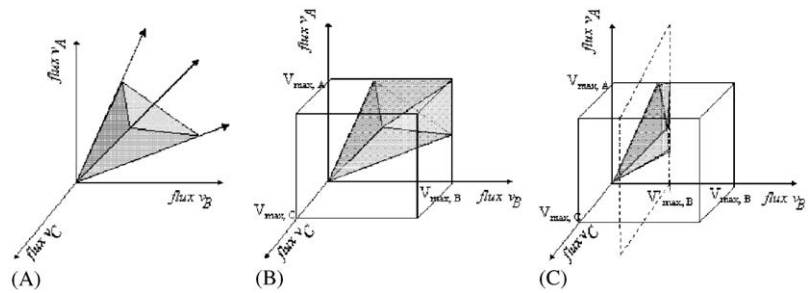


Fig. 1. Reducing the flux space. (A) The unbounded flux cone as defined by the stoichiometric matrix and irreversibility constraints. (B) The bounded flux polytope that has been capped off by the V_{max} of the reactions. The cone is surrounded by the 'bounding box' in which the Monte Carlo sampling was preformed (C) The reduced flux polytope in which $V_{max,B}$ has been significantly reduced to $V'_{max,B}$. Reducing the $V_{max,b}$ parameter 'slices off' a large portion of the flux space as shown by the cutting plane. The bounding box remains the original size, thus as the solution space gets smaller, it becomes harder to randomly select points within the space.

each reaction. This upper limit, in turn, serves to "cap off" the steady-state solution cone, forming a bounded solution space (Fig. 1B). Changes or alterations in these V_{max} values due to single nucleotide polymorphisms (SNPs) or metabolic engineering can significantly change the size of the steady-state solution space and hence the potential metabolic capabilities of the network (Fig. 1C). Thus, a reduction in V_{max} values can effectively eliminate a large number of possible states of the network. The finite polytope defined by the stoichiometric and reaction capacity constraints can range in dimension from the order of tens (for the human red blood cell) to hundreds (for genome-scale metabolic models of microorganisms). The polytope can be represented by either its bounding half planes (H-representation) or its vertices (V-representation). The constraint-based method defines the polytope by a system of equalities and inequalities, an H-representation. To calculate the volume, current methods require that all of the vertices of the polytope be enumerated. The vertex enumeration problem is #P-hard, akin to the extreme pathway enumeration problem (Samatova et al., 2002). The extreme pathways (capped off by V_{max}) comprise a subset of the extreme vertices of the polytope (excluding those that form cycles in the metabolic network). Even if the vertices could be enumerated, the process of calculating the polytope's volume is also challenging. Various algorithms exist for calculating the exact volume of a polytope from its vertices, including triangulation methods and signed-decomposition methods (for a review see Büeler et al., 2000). There are also several software packages available for volume computation, including *lrs* (Avis, 2000), cdd (Fukuda and Prodon, 1995), and the Geometric Bounding Toolbox (a Matlab© add-on from SysBrain in Southampton, England). These programs all work for small systems (less than 10 dimensions or so) but are unable to compute volumes for larger systems due to computational limitations.

To overcome the computational difficulties associated with the exact volume calculation, Monte Carlo sampling was implemented as a way to quantify the size of the steady-state solution space with respect to a reference space. In Monte Carlo sampling, points are picked randomly from the space and the fraction inside the defined constraints is counted. Monte Carlo sampling has been applied in numerous fields and various programmable algorithms are published in (Press et al., 1992; Press and Numerical Recipes Software Firm, 1994).

The goal of this study was to determine the size and shape of the steady-state flux space. For low-dimension, sample metabolic networks the hyper-volume of the space was calculated exactly. For high-dimension systems, Monte Carlo sampling was used due to the computational limitations of exact volume calculations. The Monte Carlo sampling provided a quantitative measure of the steady-state flux space's relative volume under various reaction capacity constraints.

2. Methods

2.1. The steady-state flux space: a polytope

The steady-state flux space is a finite polytope containing all possible steady-state flux distributions as defined by

$$\mathbf{Sv} = 0$$
$$0 \leqslant v_i \leqslant V_{max,i}, \quad i = 1 \ldots n, \tag{1}$$

where $\mathbf{S}$ is an $m \times n$ stoichiometric matrix (m is the number of metabolites, n is the number of reactions), and $\mathbf{v}$ is a steady-state flux distribution. Note that the reactions (v_i) are constrained to be nonnegative so reversible reactions are separated into their forward and backward parts. The edges of the flux cone are defined by the network's extreme pathways (Schilling et al., 2000). These extreme pathways are 'capped off' by the V_{max} values for the reactions and the flux cone becomes a closed polytope. We assume that $\mathbf{S}$ has full row rank (redundant equations can easily be detected and

removed). Then, the flux polytope is an $(n-m)$-dimensional object that resides in an n-dimensional space (like a two-dimensional square sitting in three-dimensional space).

2.2. Computing the exact volume of a polytope

Computing the exact volume for large, n-dimensional polytopes is an #P-hard problem (Dyer and Frieze, 1988; Khachiyan, 1993). However, it is possible to solve systems of low dimension to examine the accuracy of the Monte Carlo approximations. The exact polytope volume calculation algorithm designed by Lawrence (1991) was implemented in Matlab (Mathworks Inc, Massachusetts). Avis' *lrs* code (Avis, 2000), was used for enumerating all vertices of the polytope for input into the Lawrence algorithm. The vertices are generated in *lrs* based on the reverse search algorithm of Avis and Fukuda (1992).

In brief, the Lawrence algorithm assumes a simple polytope with the origin as a vertex. (A polytope given as an H-representation in n-space is "simple" if every extreme point satisfies exactly n of the inequalities with equality. This assumption can be binding, such as a pyramid.) The first step is to enumerate all the vertices of the polytope, as is possible for the low-dimensional sample system using *lrs*. Since the flux polytope sits in a higher dimension the polytope must be projected into its independent variables so it is full-dimensional. For each vertex, the constraints that hold with equality are determined. For each vertex,

$$\begin{aligned} D &= |\det(A_s)| \\ w &= [A_s]^{-1}c \\ f(x) &= c'x \\ N(x) &= \frac{f(x)^n}{n!Dw_1\ldots w_n} \end{aligned} \tag{2}$$

where A_s is the matrix of constraints that hold with equality at the vertex, c is a random vector such that $c'v$ is non-constant on each edge of the polytope, x is the vector of vertex coordinates, and w_i $(i=1\ldots n)$ is the ith element of w. The volume of the polytope is:

$$vol(P) = \sum_{x\in E} N(x), \tag{3}$$

where E is the set of all extreme points of P.

For the sample system studied, *lrs* was also used directly to calculate the polytope's volume from the V-representation. *lrs* uses simplicial subdivision, which partitions the space into simplicies, finds the volume of each simplex, and sums over all simplicies. Both of these volume calculation methods were used as a benchmark to test the Monte Carlo calculations in low dimensions.

2.3. Monte Carlo sampling of the convex polytope

To determine the size of a space within some known reference space (e.g., a circle within a known square), Monte Carlo is used to select points randomly (and uniformly) in the space and determine the number of 'hits' in the sub-space versus the total number of points in the reference space. Initially the reference space was determined to be a hyper-cube surrounding the polytope whose edges were defined by each reaction's V_{max} (Fig. 1B and C). However, in higher dimensions this became problematic because the actual solution space was so small, as compared with the reference hyper-cube, that no points inside the sub-space could be found. Hence, the 'effective' V_{max} had to be calculated first (see below). The effective V_{max} defines a reference hyper-cube around the polytope of interest. However, despite using the 'effective' V_{max}, points lying inside the polytope could not be found. Thus, with the bounding box method it is not possible to calculate the absolute volume for high-dimensional systems. Yet it is possible to calculate the relative volume (i.e. due to changes in V_{max}) within some chosen reference volume. The reference space was chosen to be the original flux space. Reduction in one or more V_{max} resulted in sub-space where the volume could be calculated as a percentage of the original space.

Due to the fact that the $(n-m)$-dimensional flux polytope resides in an n-dimensional space, the first step in the Monte Carlo calculation was to determine all of the independent variables in the system. This essentially 'projects' the flux polytope into a full-dimensional object. The full-dimensional polytope was sampled uniformly over the independent variables and the dependent variables were back-calculated based on the dependencies in the stoichiometric matrix, **S**. The result was a random flux distribution, **v**. The next step was to determine if the flux distribution was valid. Back-calculating the dependent flux values often results in invalid flux distributions (i.e. negative flux values or flux values that violate the V_{max} constraints). A valid flux distribution is one that meets the criteria of Eq. (1). If **v** was a valid flux distribution it was included in the sampled set. Once a set of approximately 250 000 valid points were collected, the process of determining the fractional volumes could commence. These valid points were all part of the 'base' flux polytope. From there, extra constraints (such as a lower V_{max}) were imposed and the fraction of points meeting the new criteria determined the reduction in volume. If Q is any polytope that contains our polytope, P, we can estimate the volume of P by

$$vol(P) \approx vol(Q)\,\frac{M}{N}, \tag{4}$$

where M is the number of times a point was in P, and N is the total number of random points generated (each in

Q). We choose Q such that we can easily compute its volume. For the hyper-rectangle, $Q = [0, V_{max}]$, we have $vol(Q) = V_{max\ 1} \times V_{max\ 2} \times \cdots \times V_{max\ n}$.

Let $p = vol(P)/vol(Q)$. The variance of the estimate is a function of the points sampled and is calculated as (Press and Numerical Recipes Software Firm, 1994):

$$\text{variance} = \frac{p(1-p)}{N}. \tag{5}$$

Note that the variance is minimal when p is near 0 or near 1. Being near 0, however, is the problem we want to avoid—it means we have a very small polytope inside a large hyper-rectangle. For that reason, the error rate is measured as the ratio, variance to mean, which is $p(1-p)/N : p$ or $(1-p)/N$. This gives us the intuitive result that if we can find Q that fits P more closely, we get a better estimate.

2.4. Determining the 'effective' V_{max} constraints

Due to the random sampling nature of the Monte Carlo system, it is necessary to first determine upper and lower bounds on all the variables (fluxes). Seemingly, one could use the physiologic V_{max} of the reactions. However, for the red blood cell, no points could be found inside the flux space as the bounding box defined by the physiologic V_{max} values was too large. Hence, the effective V_{max} values had to first be calculated to narrow the bounding box. The α-spectrum method developed in (Wiback et al., 2003) was applied to the flux values to determine the maximum and minimum possible flux value for each reaction. The result was a range far smaller than each individual reaction's V_{max}. The new ranges were used to form the bounding hyper-cube and the result was a valid flux distribution for approximately every 5000 points generated for the red blood cell network.

2.5. Singular value decomposition (SVD) of the flux space

Singular value decomposition (SVD) has been used to characterize the solution space as a function of its extreme pathways (Price et al., 2003) and to compare and contrast different organisms based on their stoichiometric matrices (Famili and Palsson, 2003). Here, we apply SVD to a matrix of 100 000 randomly sampled points, v_{random} where the columns are the points inside the polytope and the rows correspond to the reactions. The matrix can be decomposed into:

$$v_{random} = U\mathbf{\Sigma}V^{\mathrm{T}},$$

where the first column of U captures the most information about the polytope and gives the principal direction, $\mathbf{\Sigma}$ is a diagonal matrix of singular values. All SVD analysis was done using Matlab (Mathworks Inc, Massachusetts).

3. Results

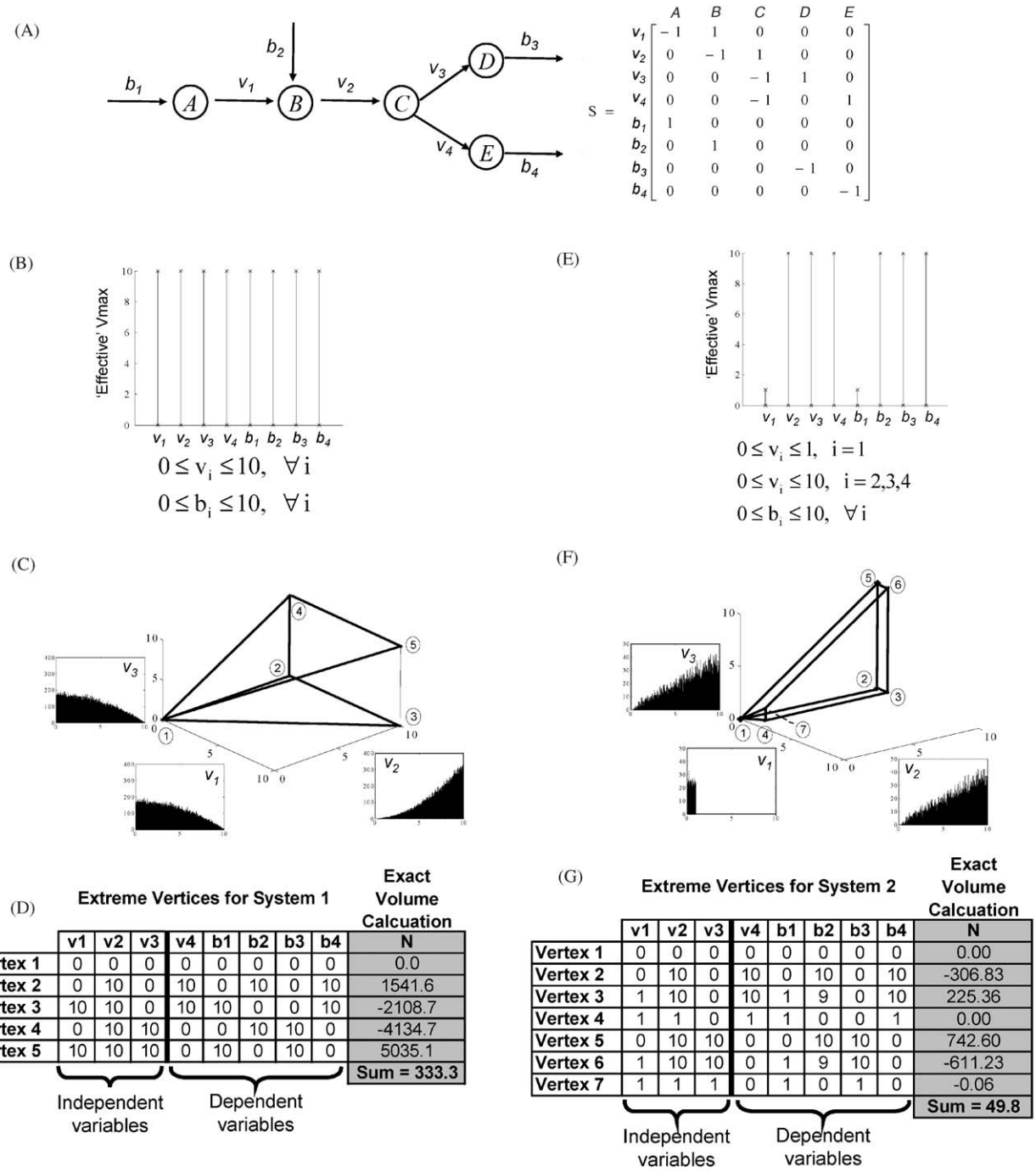
3.1. Low-dimensional sample system

For the low-dimensional sample system, the volume was determined using *lrs*, Lawerence's algorithm implemented in Matlab, and Monte Carlo sampling implemented in Matlab. For simplicity the sample system was designed such that all the exchange reactions (b_i's) were irreversible (Fig. 2A). Two cases were considered: Sample 1 in which the $V_{max}s$ on all reactions in the network were 10 and Sample 2 where the maximum capacity of reaction v_1 was lowered to 1. The examples were set up this way as the Monte Carlo method was designed to determine the relative volume of a subset to the original cone; it does not find the absolute volume. Before the Monte Carlo was preformed, the effective V_{max} for each reaction were calculated using the 'flux spectrum' method described above (Fig. 2B and C). Due to the small nature of the sample system, the effective V_{max} calculation was not necessary but is used here to illustrate its utility.

The Monte Carlo sampling does an excellent job of predicting the ratio of the two volumes (Table 1). The two polytopes are drawn in Fig. 2D and E with the independent variables (v_1, v_2, and v_3) as the axes. System 2 (Fig. 2E) is a subset of System 1 (Fig. 2D) where a large portion of System 1 was 'sliced' off by the plane formed from the $v_1 \leqslant 1$ constraint. The corresponding extreme vertices for each polytope are labeled and listed (Fig. 2F and G). Distributions of the flux values are presented for each reaction. These distributions were generated by placing the sampled points into bins based on the value for each reaction. The same set of points was used for both cases. The histograms give insight into the shape of the polytope with the greater number of sampled points falling in the wider portions of the polytope. Conversely, fewer points are found in the narrow regions of the polytope.

3.2. The human red blood cell network

The red cell network chosen was based on that used in Wiback and Palsson (2002). The network was modified to contain only irreversible reactions, used in their preferred direction (as established by the kinetic model developed in Jamshidi et al., 2001) (Fig. 3, Table 2). This modification was necessary to reduce the dimension of the Monte Carlo problem so random sampling could 'find' valid points within the space within a reasonable time period. Even with the modification, it took over a week on a Dell Dimension 8200 to generate ~250 000



	v1	v2	v3	v4	b1	b2	b3	b4	N
Vertex 1	0	0	0	0	0	0	0	0	0.0
Vertex 2	0	10	0	10	0	10	0	10	1541.6
Vertex 3	10	10	0	10	10	0	0	10	-2108.7
Vertex 4	0	10	10	0	0	10	10	0	-4134.7
Vertex 5	10	10	10	0	10	0	10	0	5035.1
									Sum = 333.3

	v1	v2	v3	v4	b1	b2	b3	b4	N
Vertex 1	0	0	0	0	0	0	0	0	0.00
Vertex 2	0	10	0	10	0	10	0	10	-306.83
Vertex 3	1	10	0	10	1	9	0	10	225.36
Vertex 4	1	1	0	1	1	0	0	1	0.00
Vertex 5	0	10	10	0	0	10	10	0	742.60
Vertex 6	1	10	10	0	1	9	10	0	-611.23
Vertex 7	1	1	1	0	1	0	1	0	-0.06
									Sum = 49.8

Fig. 2. The steady-state flux solution space of the sample network. (A) A schematic representation of the sample metabolic network and its corresponding stoichiometric matrix. (B) The flux spectrum for the sample network with 'full capacity', all fluxes can range from 0 to their V_{max} of 10. (C) The flux spectrum for the reduced capacity sample network in which $V_{max,1}$ has been reduced to 1. The flux spectrum shows that the reduction of $V_{max,1}$ also effects b_1 as both reactions can now only vary from 0 to 1. (D) A 3D projection of the reference polytope for the sample system. Distributions of flux values (histograms) of the random sampling values for each flux appear next to their corresponding axes. The circled numbers correspond to the vertex number in part (F) of the figure. (E) A 3D projection of the reduced polytope ($V_{max,1} = 1$) for the sample system. Distributions of flux values (histograms) of the random sampling values for each flux appear next to their corresponding axes. The circled numbers correspond to the vertex number in part (G) of the figure. (F) The list of vertices for the reference polytope in the sample system. The corresponding N values (as part of the exact volume computation) are given for each vertex with a random value of c. The N values sum to give the exact polytope volume. (G) The list of vertices for the reduced polytope in the sample system. The corresponding N values (as part of the exact volume computation) are given for each vertex with a random value of c. The N values sum to give the exact polytope volume.

Table 1
A summary of the results from the three methods used to calculate the volume reduction in the sample network as $V_{max,1}$ is reduced from 10 to 1. The Monte Carlo sampling produces the same result, within the error range, as the simplicial subdivision and exact volume calculation algorithms. Note the volume values are based on the 3D polytopes

		Method		
		Simplicial Subdivision (lrs)	Exact Volume Calculation (Lawrence)	Monte Carlo
3D volume	System 1 ($S1$)	$1000/3=333.3$	333.3333	106552 'hits'
	System 2 ($S2$)	$299/6=49.8$	49.8	15999 'hits'
	Ratio ($S2/S1$)	0.1494	0.1494	0.1506 ± 0.0015
	% $S2$ of $S1$	14.94	14.94	15.06 ± 0.15

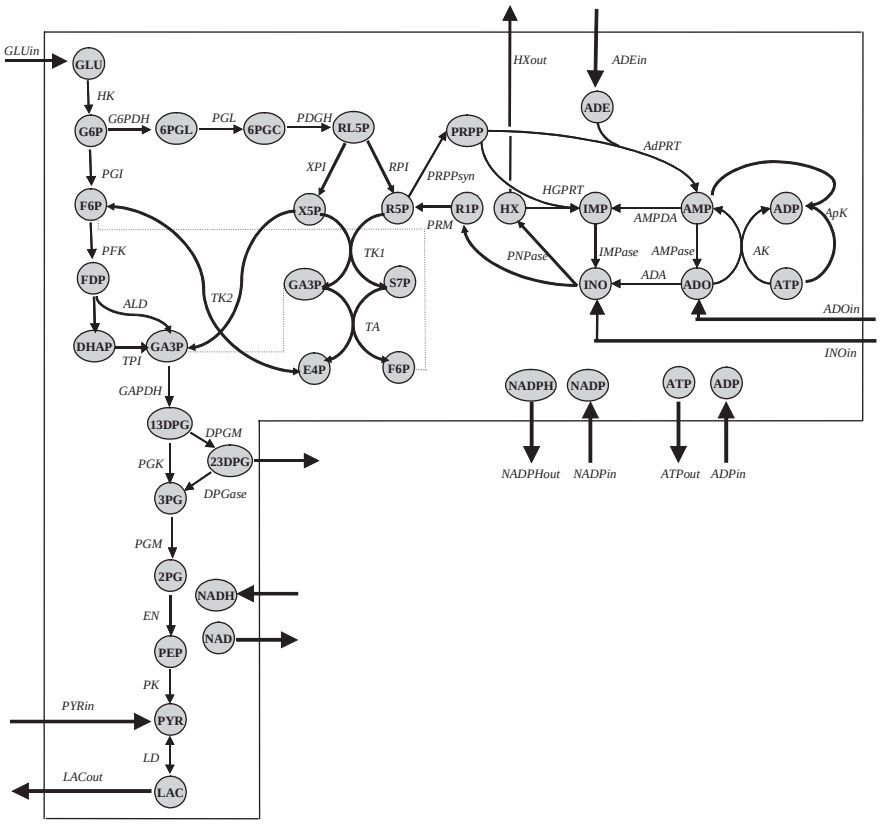


Fig. 3. A schematic representation of the human red blood cell metabolic network used in this analysis. All reactions are shown to proceed in their nominal direction.

points for the subsequent analyses. Without the modification, the polytope was too small to find with the random sampling algorithm and no valid points could be found.

3.3. The red blood 'effective' V_{max} constraints

The 'effective' V_{max} constraints for the red blood cell network were calculated based on the V_{max} values listed in (Wiback et al., 2003) (Fig. 4). For reactions where the V_{max} is unknown, the rate was set to an arbitrarily high value of 500. Note that the only reactions that truly ranged from zero to their stated V_{max} were *DPGase, AK, HGPRT, LACout, ADEin, ADOin,* and *INOin*. These reactions have some of the smallest V_{max} values and limit the rest of the network's capacity.

3.4. Distributions of flux values for individual reactions

Flux distributions for each reaction in the red cell network were plotted as histograms (Fig. 5). Several different trends appear in the distribution analysis which correspond to the shape of the polytope. The x-axis scales are different for each flux and approximately correspond to the flux spectrum ranges in Fig. 4. Note, however, that a few fluxes in the Monte Carlo

Table 2
The stoichiometric matrix of the human red blood cell with all reactions irreversible (set to proceed in the 'nominal' direction)

S =	G6P	F6P	FDP	DHAP	GA3P	13DPG	3PG	2PG	PEP	GL6P	GO6P	RU5P	R5P	Xu5P	S7P	E4P	PRPP	IMP	R1P	AMP	GLC	23DPG	PYR	LAC	HX	ADE	ADO	INO	ADP	ATP	NAD	NADH	NADP	NADPH
HK	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	1	-1	0	0	0	0
PGI	-1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
PFK	0	-1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	-1	0	0	0	0
ALD	0	0	-1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TPI	0	0	0	-1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
GAPDH	0	0	0	0	-1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	1	0	0
PGK	0	0	0	0	0	-1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	1	0	0	0	0
DPGM	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
DPGase	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0
PGM	0	0	0	0	0	0	-1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
EN	0	0	0	0	0	0	0	-1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
PK	0	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	-1	1	0	0	0	0
LDH	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	1	0	0	0	0	0	0	1	-1	0	0
G6PDH	-1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	1
PGL	0	0	0	0	0	0	0	0	0	-1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
GL6PDH	0	0	0	0	0	0	0	0	0	0	-1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	1
RPI	0	0	0	0	0	0	0	0	0	0	0	-1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
XPI	0	0	0	0	0	0	0	0	0	0	0	-1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TKI	0	0	0	0	1	0	0	0	0	0	0	0	-1	-1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TKII	0	1	0	0	1	0	0	0	0	0	0	0	0	-1	0	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TA	0	1	0	0	-1	0	0	0	0	0	0	0	0	0	-1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
AMPase	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	1	0	0	0	0	0	0	0
ADA	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	1	0	0	0	0	0	0
AK	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	-1	0	1	-1	0	0	0	0
ApKR	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	2	-1	0	0	0	0
AMPDA	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
AdPRT	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	1	0	0	0	0	0	-1	0	0	0	0	0	0	0	0
IMPase	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
PNPase	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	-1	0	0	0	0	0	0
PRM	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
PRPPsyn	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0
HGPRT	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	1	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0
GLCin	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
23DPGout	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0	0
PYRout	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	0
LACout	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0	0
Hxout	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	0	0	0	0	0	0	0
ADEin	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
ADOin	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
INOin	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
ADPin	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
ATPout	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	0	0
NADin	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
NADHout	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0
NADPin	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
NADPHout	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1

simulation did not reach their full 'potential'. In other words, many of the reactions in glycolysis and the pentose phosphate pathway did not cover the full range of possible flux values in the random sampling. The smaller range is most likely due to the fact that the flux polytope is very narrow in these places (as evidenced by the distributions that trail off at the higher values) and hence were missed by the Monte Carlo sampling procedure.

The flux values for the nominal steady state of the red blood cell, as calculated with the kinetic model from (Jamshidi et al., 2001), are marked with the arrows in Fig. 5. All the reactions fall well within the bounds of the polytope. In addition, 'lethal' stressed states calculated with the model (Jamshidi et al., 2002) also appear to fall within the bounds of the flux ranges despite the cell's in silico 'death' (results not shown). The fact that the cell cannot sustain these loads in silico yet do not appear to hit the applied constraints points to the fact that there are additional constraints placed on the cell that are not captured by the stoichiometry and reaction capacities. These constraints could include kinetic limitations, osmotic balances, and concentration capacities. Incorporating these constraints will further shrink the solution space.

3.5. Varying the V_{max} values individually

The V_{max} values were varied individually to look at their impact on the size of the solution space. Each V_{max} was reduced to 50% and 10% of its nominal value and the size of the reduced solution space was compared to the original (Fig. 6). In some cases, the size of the polytope is reported as 0, which means no points could be found inside the reduced space. The polytope still exists but it is just too small a sub-space to detect with the random sampling. Reactions that respond identically (reaction subsets as defined in Schilling and Palsson, 2000) are lumped on the y-axis in Fig. 6. The histograms in Fig. 5 can be used to interpret the V_{max} reduction results in Fig. 6. The reactions with 'normal' distribution shaped histograms such as HK, PFK, and PRM show little reduction in the size of the space when reduced 10% (as it only cuts off the small tail) but huge reduction when reduced to 90% (as the main bell part of the curve is cut off and only the small leading edge

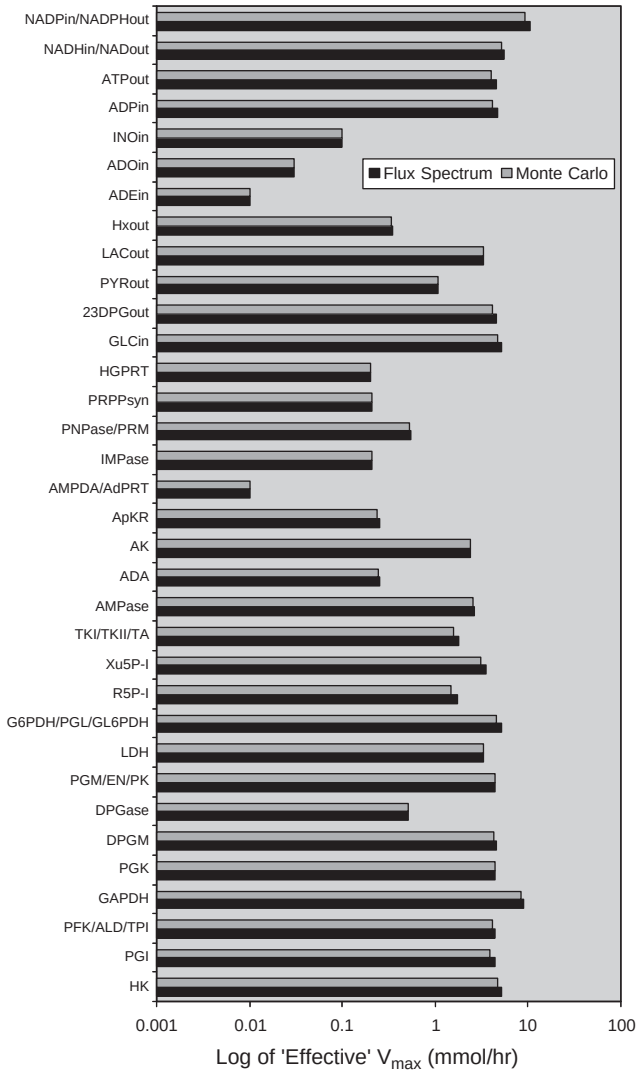


Fig. 4. The flux spectrum versus the Monte Carlo spectrum. The black bars show the flux spectrum for the red blood cell. The gray bars show the range of values for each flux as calculated using the Monte Carlo sampling. The V_{max} values were plotted on a log scale. In some cases, the Monte Carlo sampling falls short of the effective V_{max}, meaning that part of the polytope was too narrow or small to sample.

remains). Reactions such as R5P and PGI in which the majority of the fluxes appear at the low end of the histogram do not affect the size of the polytope as significantly.

3.6. SVD of the flux polytope

The SVD of the flux polytope was calculated under various constraints to determine the effect of uptake and secretion rate knowledge on the size of the solution space. The nominal steady-state flux ($\mathbf{v}_{ss}$) distribution for the red cell was calculated using the kinetic model developed in Mathematica® (Jamshidi et al., 2001). This flux vector was used as the reference vector for determining how close the first mode of the SVD was to the physiologic steady-state solution. The angle between the nominal steady-state vector ($\mathbf{v}_{ss}$) and the first mode of the SVD (U_1) was calculated as

$$angle = \cos^{-1}\left(\frac{\mathbf{v}_{ss}U_1}{|\mathbf{v}_{ss}||U_1|}\right).$$

The resulting angle was plotted on a polar coordinate system (Fig. 7). For the entire flux cone (100% of the original volume), the first mode of the SVD was 30° from the nominal steady-state vector. The size of the steady-state flux cone was reduced by adding constraints on the uptake and secretion rates (since, experimentally, these are the most easily measurable fluxes). The selected fluxes in Fig. 7 (glucose input, 23DPG output, and lactate output) were fixed to $\pm 10\%$ of their nominal value individually and together and the size of the flux cone was recalculated (i.e. all points falling outside the constraints were eliminated). As the constraints on the cone tighten the size of the cone decreases and first mode of the SVD moves closer to the steady-state vector (i.e. the cone closes in on the 'true' solution).

4. Discussion

The work presented herein represents the first attempt at quantifying the size and shape of the steady-state flux solution space. It was possible to calculate the exact volume of the flux polytope for dimensions less than 10 and thus determine key features of the space such as size, shape, and parametric sensitivities. For spaces of larger dimensions, computational limitations prevent the exact volume calculation. As an alternative, a Monte Carlo sampling algorithm was used to determine the relative size of the flux polytope in these higher dimensions. Relative size calculations are used to determine the effects of parameter variations. Further, Monte Carlo sampling allows one to determine the statistical properties of the distribution values for any given variable in the space. The statistical properties are readily re-evaluated is the space is reduced in size by the imposition of additional constraints or by varying parameter values.

The Monte Carlo sampling method was validated on a low-dimensional system for which the exact polytope volume could be calculated using simplicial subdivision (*lrs*) and signed decomposition (Lawrence's method). Distributions were plotted of the random flux values for each reaction which gave insight into the shape of the polytope. The flux distributions were predictive of the reduction in size of the metabolic solution space by decreasing a reaction's V_{max}.

The Monte Carlo sampling method was applied to the red cell metabolic network to determine the relative size

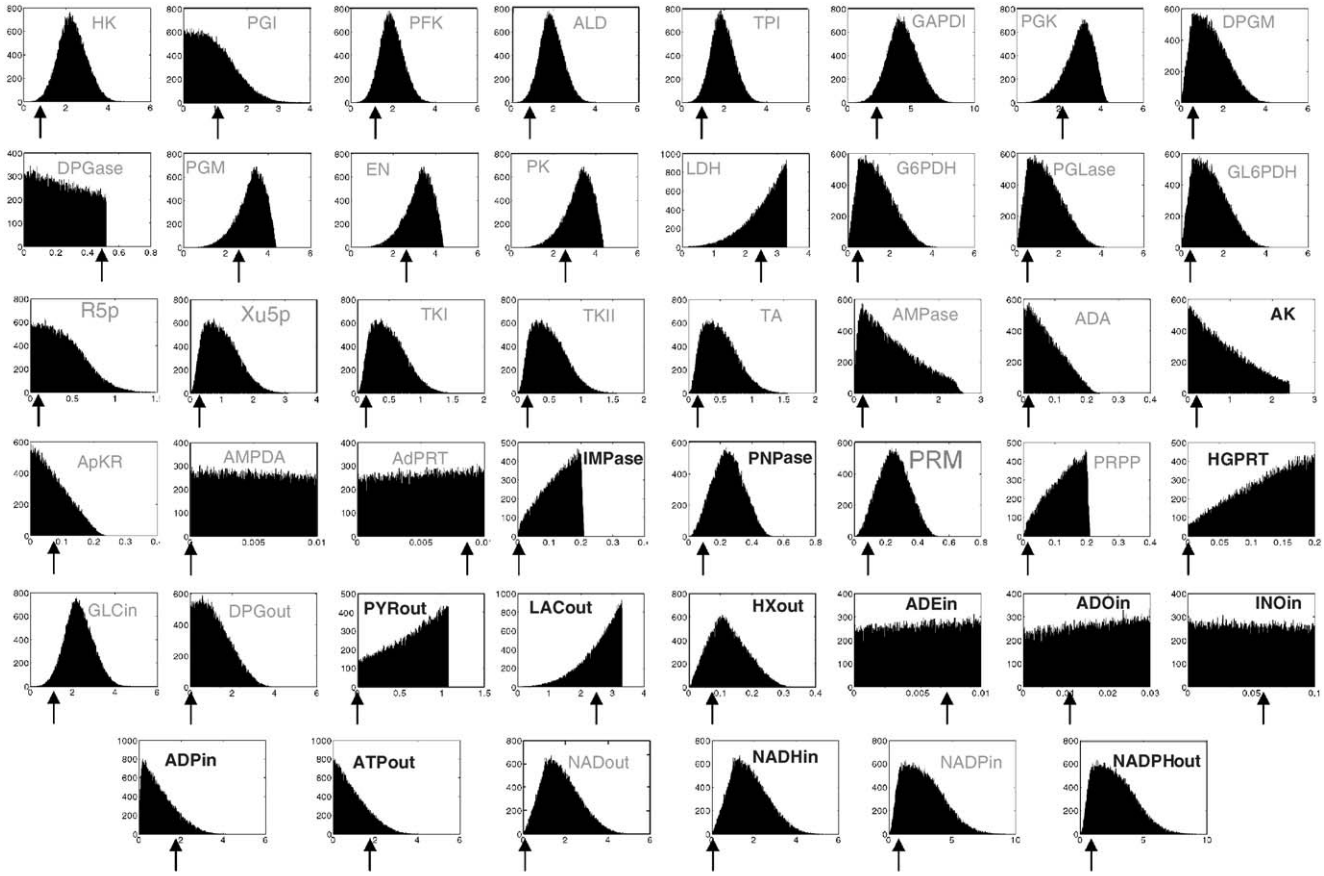


Fig. 5. Distributions of flux values for each of the 46 reactions in the red cell network showing the distribution of points in the Monte Carlo sampling. The flux names in bold face are the independent variables while the gray flux names are all dependent variables. The arrows point to the nominal steady-state flux value for each reaction as calculated from the kinetic model (Jamshidi et al., 2001).

of its steady-state solution space. It was shown that each reaction V_{max} has a vastly different effect on the size of the solution space. The size reduction in the flux cone varied from 7% to 96% depending on which reaction's V_{max} was decreased by 10%. It was also shown that the Monte Carlo sampling method 'missed' some of the narrower regions of the flux space, failing to sample over the entire range of effective V_{max} values. The nominal steady-state flux distribution was shown to fall well within the limits of the steady-state flux cone as illustrated in Fig. 5. For a known flux distribution in the red cell, SVD was used to evaluate the effect of various constraints on the shrinkage of the solution space. The effect of the constraints was measured as the angle between the known distribution and the first mode of the SVD of the matrix of randomly sampled points inside the polytope. The addition of experimentally measurable constraints of the glucose uptake rate and 23DPG and lactate secretion rates significantly shrunk the size of the cone and moved the first mode of the SVD much closer to the nominal steady state.

The general Monte Carlo sampling method is highly inefficient for calculating the volume of a high dimensional polytope. The highly conservative bounding box as defined by the effect V_{max} values resulted in only one 'valid' flux distribution for every 5000 random points generated. Methods are currently being developed that narrow the bounding box to a tighter fitting parallelepiped so that a higher fraction of random polytopes can be found (Schellenberger and Price, pers. comm.). Currently, the bounding box was too large to allow for the detection of the flux polytope so only a comparative volume could be calculated as some fraction of a reference polytope. With the parallelepiped method, the volume of the parallelepiped is known and the fraction of points within the flux cone can be calculated which leads to the absolute volume. In addition the parallelepiped method significantly lessens the amount of time needed sample the space—reducing it from days to minutes in initial test cases with the red cell system (Schellenberger and Price, pers. comm.).

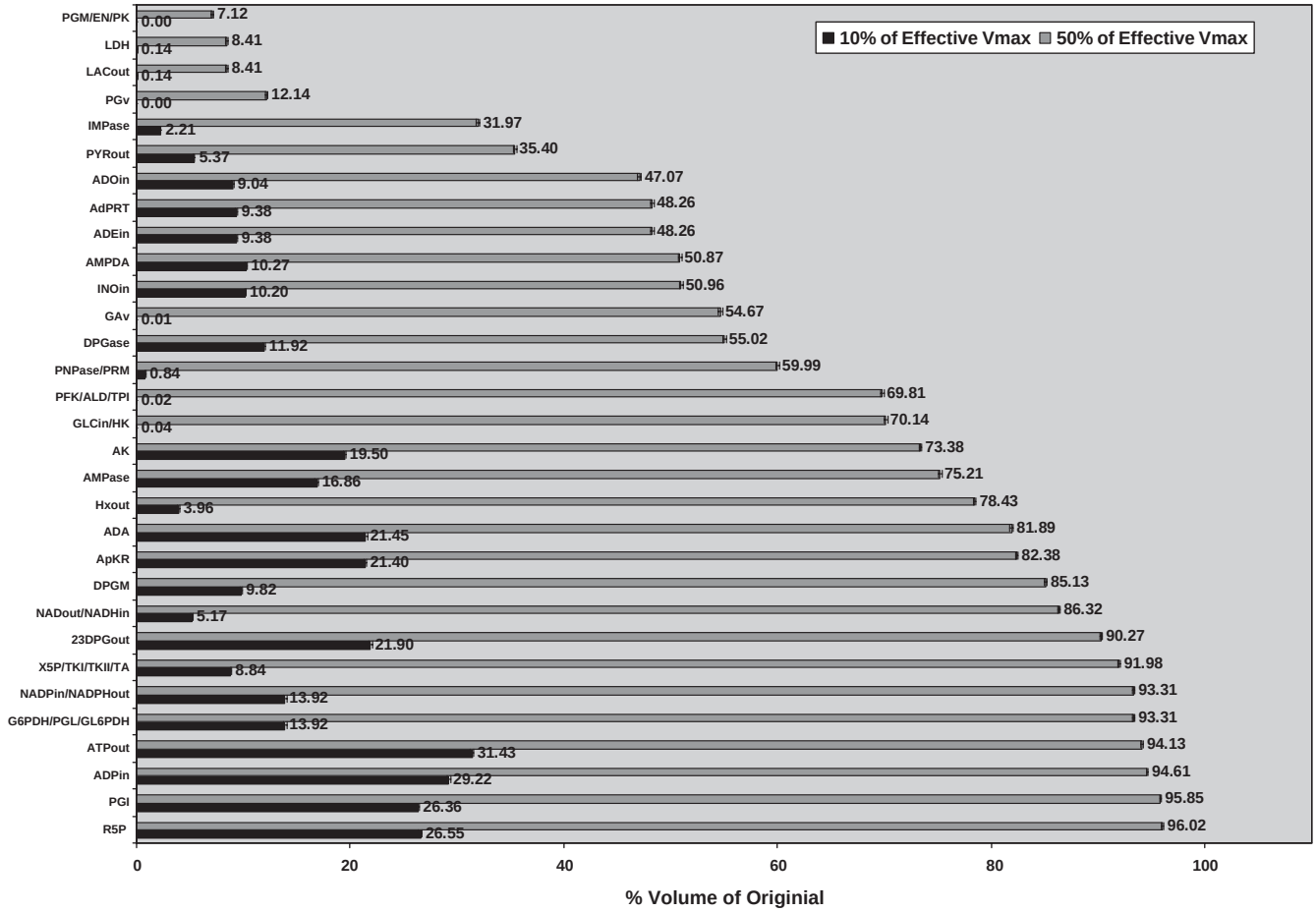


Fig. 6. The effect of V_{max} reduction on the size of the flux polytope. The gray bars correspond to the percent reduction in the hyper-volume of the flux polytope when each reaction V_{max} is reduced by 50%. The black bars are the hyper-volume when each reaction V_{max} is reduced by 90%. Fluxes for which the reduction is identical (i.e. they are a reaction subset Schilling and Palsson, 2000), are lumped on the y-axis.

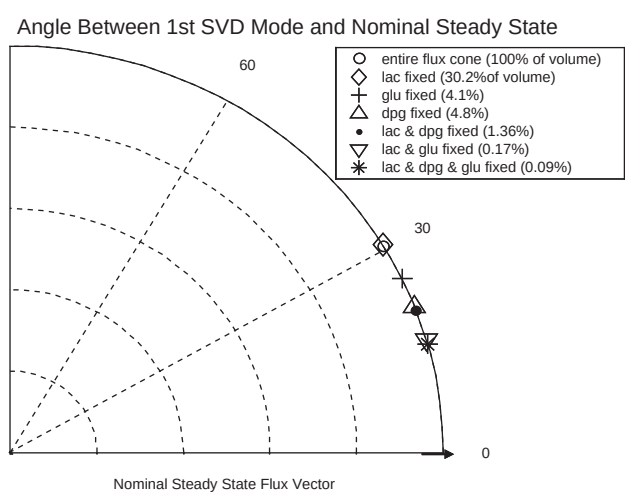


Fig. 7. A polar plot of the angle between the first mode of the SVD for different polytopes and the nominal steady-state flux vector. Each point represents a different set of constraints that were placed on the solution space. The SVD of the random points in the space was calculated and the angle between the first mode of the SVD and the nominal steady-state vector was plotted. The volume for each constraint set is included in the legend.

Being able to determine the size of the steady-state flux space should prove useful in determining the effect of adding more constraints to the system. The constraints can be evaluated based on the amount of reduction that is achieved in the flux polytope. Similar work is currently being done with regards to the concentration polytope and its reduction with kinetic constraints (Famili, pers. comm.). The effect of single nucleotide polymorphism (SNPs) on the system could also be studied as such enzyme defects often lower the reaction's effective V_{max}. In this case, it is important to note that it is the shrinkage of the space that contains the physiological steady state that matters. If a constraint eliminates a large section of the solution space that does not contain the cell's operating point, than there will probably be no effect on the cell's function. However, if a constraint eliminates a section of the flux cone that contains the physiologic state then the cell's operating point will most probably be affected, as has been demonstrate with the in silico study of glucose-6-phosphate dehydrogenase and pyruvate kinase SNPs in (Jamshidi et al., 2002).

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The reader might want to be aware of a related study where the entire solution space of a genome-scale model of *E. coli* was sampled using a hit and run sampling method (Almaas et al., 2004).

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